

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and executed small-scale protein purification projects using **AKTA systems** and various **chromatography techniques**, delivering scientific studies with limited supervision.
- Developed and maintained expertise in protein purification methodologies, authoring comprehensive **SOPs**, **risk assessments**, and technical reports to a high standard.
- Supported root cause analysis investigations for bioprocessing issues, troubleshooting manufacturing problems and enhancing technical depth for junior scientists.
- Managed cross-functional project plans with peers in Process Development and Analytical Services, ensuring alignment with the Biological Centre of Excellence (BCoE) business objectives.
- Authored protocols and work instructions, maintaining a high standard of **Good Laboratory Practices (GLP)** and contributing to a culture of operational excellence.
- Presented scientific output and project findings to internal functions and BCoE stakeholders, promoting the work of the BCoE across diverse technical teams.

SKILLS

Bioprocessing Techniques – Small-scale protein purification, AKTA systems, TFF, Filtration, Chromatography, Resin screening

Scientific Documentation – SOPs, Protocols, Work Instructions, Risk Assessments, Technical Reports, GLP

Data Analysis & BI – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn

Software Engineering – Java, Spring Boot, Python, REST APIs, SQL, PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS, Docker, Jenkins, Git, Linux

Quality & Compliance – GMP, ICH guidelines, Change Management, Regulatory Awareness

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Bioprocess Optimization Dashboard

- Developed an interactive dashboard using **Power BI** to visualize and analyze data from small-scale protein purification runs, enabling faster identification of process deviations.
- Integrated data from various sources including experimental results and equipment logs, applying **DAX** measures for critical KPI calculations related to yield and purity.

- Published curated reports to business stakeholders, facilitating data-driven decision-making for process improvement initiatives within the R&D; team.

Automated Data Pipeline for Lab Experiments

- Built a Python-based ETL pipeline to automate the extraction, transformation, and loading of experimental data from laboratory instruments into a centralized data store.
- Implemented data validation checks and error handling routines, ensuring data integrity for downstream analysis and reporting, improving data processing efficiency by **30%**.
- Managed the pipeline using **Jenkins** for scheduled execution and monitoring, contributing to the overall reliability of the data infrastructure supporting bioprocess research.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)